

Inference as Currency

A monetary theory of the AI cycle, in three pages.

Microsoft cancelled its internal Claude Code seats this November. The cost-benefit math didn't clear. Microsoft is the most-aligned buyer for token-priced AI services that exists — owns Azure, holds nineteen percent of OpenAI, sells more enterprise software than any other firm on earth. If Microsoft can't make the math work, the buyer-side problem is deeper than the literature has admitted. The same month, Uber burned its entire 2026 AI budget in four months. GitHub deprecated flat-rate Copilot. Anthropic moved from a token meter to opaque weekly caps. OpenAI walked into a Y Combinator dinner and offered two million dollars in tokens for equity in every founder in the room.

The consensus view writes much larger numbers. Trillion-dollar lab revenues by 2030, with "AI finance ministers and central bankers" driving 10% GDP growth in the developing world;² labs priced on Netflix-and-Spotify comparables, durable margins, personalisation moat, commoditisation thesis "more or less irrelevant";³ an \$800B market the lab's seed investors call "a gross underestimate."¹ None of those projections survives the November data.

The platform-software comparables are wrong. AI labs are private currency issuers. The currency is the token. The underlying asset is the *verifier substrate* — the human or software that decides whether AI work was done correctly. Token value behaves like nineteenth-century private banknote value: face inside the issuer's integrated perimeter, discounted with distance, vulnerable to bank runs when redemption fails. The rent on a token is set by the K-complexity of verification in the buyer's domain — not by the technical capability of the model, not by the volume of tokens issued. The labs' visible behaviour is the rational response of private banks to redemption stress on their notes.

The token

A token is a claim on inference, redeemable for verified economic value. The verification happens through a substrate: a compiler accepts the code, a partner signs the brief, a clinician signs the chart, a regulator approves the filing.

Kolmogorov complexity of verification, $K(\text{verify})$, is the length of the shortest program that maps outputs to yes-or-no judgments given the verifier's information set.⁴ For a compiler, $K(\text{verify})$ is small — language semantics plus a test harness. For a partner reviewing a brief, $K(\text{verify})$ is enormous — the assigned judge's history, opposing counsel's playbook, the client's risk tolerance, the firm's strategy. Most of it tacit.

The rent on a token tracks $K(\text{verify})$, not model capability. Where verifiers are cheap (code, formal math, schema-deterministic audit), the rent flips to whoever owns the compute. Where verifiers are humans, the rent stays on the signature. Tokens are derivative of verifiers.

Discount by distance

A token from Anthropic redeems at face inside Claude Code. The same token used in a third-party application without Anthropic's vertical integration redeems at a discount — the buyer has to construct her own redemption infrastructure. The same token used in a generic domain without a productisable verifier doesn't redeem at all.

The pattern is nineteenth-century banknote economics.⁵ A note from a bank in Indiana traded at face inside the bank's home county, at a discount in the next state, and as wallpaper across the Mississippi. Free Banking America (1837–1863) produced 800+ private currency issuers. The eventual resolution was the National Banking Acts of 1863–1864 — unified federal currency, taxes that killed state banknotes, federally chartered banks.

Formally:

$$V(d) = V_0 \cdot e^{-\lambda d}$$

where d is the buyer's institutional distance from the lab's perimeter and λ is the lab-specific discount rate, set by perimeter investment. The lab's strategic problem is to keep λ low across the widest possible D .⁶ Two corner solutions: deepen the perimeter (vertical integration), or extend it (subsidised circulation).

Two strategies

Anthropic runs perimeter deepening. Direct enterprise contracts in regulated verticals — clinical decision support, claims adjudication, regulatory affairs — account for roughly 80% of revenue. First-party harnesses (Claude Code, Coefficient) capture token volume inside the lab. The OAuth restriction on third-party harnesses pulls telemetry inside the perimeter. Compliance infrastructure supports buyers in regulated industries. The lab is becoming the vertical SaaS company in domains where the model is one component of a stack it owns end-to-end. The bet: $K(\text{verify})$ stays high; the residual rent accrues to substrate owners.

OpenAI runs perimeter extension.⁷ Multi-vendor supply agreements with Microsoft, Amazon, and Google flatten hyper-scaler dependence into a many-counterparty asymmetry. A horizontal API business prioritises developer adoption over

vertical depth. The \$2M YC token offer buys equity in tomorrow's app businesses paid in OpenAI's own currency. The startups burn the tokens building products that route inference through OpenAI. The lab captures the equity upside *and* the metered revenue. The bet: $K(\text{verify})$ falls; volume across many small substrates dominates depth in any one.

Both strategies are autarky. Each reduces exposure to external parties whose decisions could break the lab's operations. The labs are not competing for the same customer. They are betting on opposite futures of the same parameter.

Bank runs

The Microsoft cancellation is a single-buyer redemption failure — a depositor who showed up for specie and was politely turned away. The bank survives the single withdrawal. The information about whether the bank could honour at face propagates. Other large enterprises run their own cost-benefit math against the same flat-rate subscription. Enough simultaneous cancellations and the lab faces a coordinated run.

Stablecoin history is the modern training set.⁸ TerraUSD destroyed \$45 billion in a week when its algorithmic peg failed. USDC depegged in March 2023 when Silicon Valley Bank failed. Tether's reserves were 27.6% of issued tokens for two and a half years. The lesson: private currency without sufficient backing fails not gradually but suddenly, when buyer confidence cracks.

Anthropic's weekly rate caps and OpenAI's priority-service contracts are tacit admissions that the per-token meter is not a stable medium of exchange. Both labs are converting their banknotes into bilateral capacity reservations — the retreat Free Banking-era banks made when they ran out of specie.

Resolution

Three paths.

State imposition. The federal government declares the inference economy strategic, imposes a regulated-issuer framework, perhaps absorbs the labs into a Manhattan-Project-style consortium.⁹ Mechanism: the National Banking Acts of 1863–1864. Triggers: a visible bank-run event, geopolitical pressure, political will. Two of three are partially formed.

Serial failure. The British nineteenth-century banking arc. Smaller labs fail or merge into larger ones. Larger labs experience their own time-gap stress and compress to defense-contractor multiples. A stable oligopoly settles with smaller revenue bases than current equity valuations imply.

Multipolar fragmentation. The US lab system persists alongside Chinese, European, Indian, and Middle Eastern parallel systems. Token convertibility across borders becomes a regulated activity. The most powerful institutions are not the labs but the cross-border verifier-credentialing bodies — Underwriters Laboratories, the AICPA, Moody's analogs in each jurisdiction.

Probability is roughly equal across the three on a five-year horizon. The realised outcome will be a combination.

What this means

Lab equity is priced on platform-software comparables. The realised revenue and margin look like cost-pass-through infrastructure. The implied multiple compression — from platform multiples to monetary-institution multiples — is the modal expected outcome, not a tail risk.

The institutions that benefit are verifier-substrate owners. Credentialing bodies. Audit firms. Vertical software companies that own the substrate in their domain — Halter, Waymo, OpenEvidence, Sardine, Abridge. They trade at infrastructure multiples and they will earn them. Hardware and energy infrastructure benefits regardless of which resolution path obtains.

The credentialed human at the bottom of the signature chain is the closest thing the AI cycle has produced to specie reserves. She holds value across every redemption scenario. No lab can capture her without integrating into the substrate where her signature lives. The wage of the doctor, the lawyer, the engineer, the underwriter, the pilot rises in real terms over the next five years. The middle of the labour market — the analyst, the coordinator, the entry-level associate, the role whose function was to assemble inputs for someone else's signature — compresses out of every column simultaneously. The political debate about AI and labour has been sighting the wrong tier.

The cycle has been pricing the wrong substrate. The repricing comes either through state intervention, through consolidation, or through fragmentation. All three are visible in the current data.

1. Anjney Midha (Anthropic seed investor, ex-a16z): "in the early days, a couple Anthropic cofounders and I started doing weekly working sessions for our first fundraiser we needed a coherent view of our market size and arrived at an answer that felt reasonable: ~\$800B/year as Andrej has discovered, that was a gross underestimate" — [X.com](#), Oct 2025. Dwarkesh Patel adds that founders and early investors of frontier labs will "own a galaxy."

2. Dario Amodei, "Machines of Loving Grace" (Oct 2024). Targets 20% annual GDP growth in the developing world, with AI replicating East Asian-style ~10% growth via "AI finance ministers and central bankers." Trillion-dollar revenues by 2030 are presented as plausible under the bull-case capability trajectory.

3. Dan Sundheim, D1 Capital, Invest Like the Best Ep. 460 (Feb 2026): LLM businesses combine fixed-asset economics (Netflix content) with personalisation-driven stickiness (Spotify playlists); gross margins are durable; the commoditisation thesis “is more or less irrelevant.” The strongest published counter to the framing here.
4. The K(verify) construction extends conditional Kolmogorov complexity (Li & Vitányi, *An Introduction to Kolmogorov Complexity*, 4th ed., Springer 2019) using a verifier’s tacit information set as the conditioning variable. Underlying economic machinery: Akerlof, “Lemons” (*QJE* 1970); Holmström, “Moral Hazard and Observability” (*Bell J. Econ.* 1979); Bergemann & Bonatti, “Markets for Information” (*Annu. Rev. Econ.* 2019).
5. Hugh Rockoff, “The Free Banking Era: A Reexamination,” *Journal of Money, Credit and Banking* 6:2 (1974), pp. 141–167. For the political economy of the resolution, Calomiris & Haber, *Fragile by Design* (Princeton, 2014). The discount-by-distance pattern was geometric in observable broker quotes by 1850.
6. Ronald Coase, “Durability and Monopoly,” *J. Law & Econ.* 15 (1972). Board & Pycia, “Outside Options and the Failure of the Coase Conjecture,” *American Economic Review* 104:2 (Feb 2014). The labs’ eighty-percent gross margins are the Board–Pycia selection signature, not a moat.
7. Joel Spolsky, “Strategy Letter V: Commoditize Your Complement,” joelonsoftware.com (June 2002). Each layer’s complement is what it tries to cheapen. Hamilton Helmer, *Seven Powers* (2016), on counter-positioning: the labs’ token meter is counter-positioned against the SaaS flat-rate model, and SaaS cannot retrofit per-token pricing without breaking customer relationships.
8. TerraUSD / LUNA collapse, May 2022 — algorithmic peg failed; ~\$45B value destroyed in a week. USDC depeg, March 2023 — Silicon Valley Bank held Circle’s reserves. Tether reserve fraction during 2016–18 was 27.6% per the NY Attorney General’s 2021 settlement. Three independent demonstrations that private currencies without sufficient backing fail discontinuously when confidence cracks.
9. Leopold Aschenbrenner, “Situational Awareness: The Decade Ahead” (June 2024), section “The Project”. Predicts USG absorption of frontier labs into a Manhattan-Project-style consortium by 2027–28 under Lockheed/Boeing-type defense contracting arrangements, triggered by capability + geopolitical pressure.